

Topic:-

Prepare a list of the chemical changes that occur in your house and surroundings and discuss them in the class.

Introduction:-

Changes are taking place constantly all around us in the physical universe. Ice cubes melt into liquid water on a warm table, glass tumblers shatter into pieces if dropped, green raw mangoes ripen into sweet yellow fruit, fresh milk turns into thick sour curd, and shiny iron bicycle chains develop a crust of reddish-brown rust during the monsoon.

In science, all transformations are categorized into two fundamental types:

Physical Changes and Chemical Changes. A physical change alters only the physical state, shape, or size of matter without altering its internal molecular composition, and it can usually be reversed (like freezing water into ice). In contrast, a chemical change is a permanent transformation in which original chemical bonds are broken and completely new substances with entirely different physical and chemical properties are formed. In this project, we surveyed our kitchen, garden, and neighborhood to identify and catalog everyday chemical changes.

❖ Key Signs That a Chemical Change Has Occurred:-

- Emission of heat, light, or sparks (like burning of cooking gas or matchsticks).
- Irreversible change of color (like iron nails rusting reddish-brown).
- Evolution of a new gas with bubbling or characteristic odor (like curdling milk).
- Formation of a permanent insoluble precipitate that cannot be reversed.

Aim & Household Survey Plan:-

Aim: To observe, record, and prepare an exhaustive catalog of chemical changes occurring in our household, kitchen, garden, and neighborhood, and lead a discussion in our science class.

❖ Objectives of the Project:-

- To identify and distinguish between physical and chemical changes in everyday life.
- To write simple word equations for common chemical transformations at home.
- To understand how to prevent harmful chemical changes like rusting and food spoilage.

❖ Household Areas Surveyed:-

- **Kitchen:** Cooking food, baking chapatis, milk curdling, fruit ripening, boiling eggs.
- **Garden & Balcony:** Composting fallen dry leaves, rusting iron window grills.
- **Surroundings:** Burning of LPG gas, hardening of wet cement during construction.



Chemical Changes Observed at Home:-

Our home survey documented several fascinating chemical reactions taking place daily:

- **1. Turning Milk into Curd (Fermentation):** Warm milk mixed with a spoonful of starter curd undergoes biological fermentation. Helpful *Lactobacillus* bacteria convert milk lactose sugar into lactic acid. The increasing acidity coagulates casein proteins, turning liquid milk into thick, delicious sour curd. You can never turn curd back into milk!
- **2. Rusting of Iron Grills (Oxidation):** Iron railings, nails, and bicycle chains exposed to moist air react with atmospheric oxygen and water vapor: $4\text{Fe} + 3\text{O}_2 + 2x\text{H}_2\text{O} \rightarrow 2\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ (hydrated iron oxide). Rust is porous, flaky, and eats away the metal strength.
- **3. Combustion of Cooking Gas (LPG):** Turning on the gas stove burns butane and propane hydrocarbons in atmospheric oxygen: $\text{C}_4\text{H}_{10} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{Heat}$, producing clean blue fire for cooking.
- **4. Enzymatic Browning of Cut Apples:** When an apple is sliced, polyphenol oxidase enzymes inside cells react with air, oxidizing plant tissues into brown melanin pigment within 10 minutes.

Comprehensive Survey Data Table:-

The table below logs the common changes observed across our home and surroundings:

<i>Observed Activity</i>	<i>Initial Substances</i>	<i>New Product Formed</i>	<i>Evidence of Chemical Change</i>	<i>Reversibility</i>
Making Curd	Liquid milk + curd starter	Thick sour curd (Lactic acid)	Sour taste, thick solid texture	Irreversible
Rusting of Iron	Shiny grey iron metal (Fe)	Reddish-brown rust flakes	Color change, crumbly decay	Irreversible
LPG Stove Burning	Butane gas + Oxygen	Carbon dioxide + Water vapor	Blue flame, intense cooking heat	Irreversible
Boiling an Egg	Liquid egg white & yolk	Solid rubbery egg protein	Heat denatures protein chains	Irreversible
Ripening of Mango	Hard green sour fruit	Soft sweet yellow mango	Color turns yellow; sweet smell	Irreversible
Composting Leaves	Dry garden leaves	Rich black humus organic manure	Decay heat, dark earthy smell	Irreversible

Scientific Analysis & Discussion:-

Our classroom discussion explored the atomic and molecular science behind these changes:

- **Breaking and Making of Chemical Bonds:** In physical changes (like melting ice or tearing paper), the chemical bonds between atoms remain intact— H_2O remains H_2O whether solid or liquid. In a chemical change, old covalent or ionic bonds break apart and atoms rearrange into entirely new molecular configurations.
- **Beneficial vs Harmful Chemical Changes:** We categorized chemical changes into two groups:
 - *Beneficial Chemical Changes:* Photosynthesis in trees, cooking food, curdling milk, composting waste, and digestion inside our stomach.
 - *Harmful Chemical Changes:* Rusting of bridges and gates, spoiling of cooked milk, decaying of teeth, and rancidity of cooking oils.
- **Energy Flow:** Burning LPG is exothermic (releases heat), while baking a cake in an oven is endothermic (absorbs heat to cook starch).

Everyday Prevention & Protective Methods:-

Knowing chemical science helps us control and prevent unwanted chemical reactions:

- **1. Preventing Iron Rusting:**

- *Painting:* Applying oil paint or primer onto iron window grills seals the metal surface, stopping oxygen and moisture contact completely.
- *Galvanization:* Coating iron with a thin layer of zinc metal protects municipal water pipes from rusting.
- *Oiling:* Applying grease to bicycle chains reduces friction and stops rust.

- **2. Preventing Food Spoilage:** Storing cooked food inside cold refrigerators (below 4°C) slows down microbial chemical reactions by 90%, keeping food fresh for days.

- **3. Stopping Apple Browning:** Squeezing a drop of lemon juice (vitamin C antioxidant) over cut apple slices prevents oxygen from browning the fruit!

What We Learned & Conclusion:-

- A chemical change permanently forms new substances with completely different properties by rearranging chemical bonds.
- Chemical changes are irreversible and accompanied by energy exchange, color change, or gas evolution.
- Cooking, fermentation, respiration, and rusting are ubiquitous chemical reactions occurring daily around us.
- Applying protective coatings (painting, oiling) and refrigeration effectively controls unwanted chemical degradation.

Conclusion:-

Preparing this catalog of chemical changes across our household and neighborhood opened our eyes to the constant molecular transformations that shape our everyday world. From the kitchen stove to our own body's digestion, life is powered by chemical science. Discussing our findings in class helped all students understand that chemistry is not confined to test tubes—it is actively unfolding in every corner of our daily lives.

References:-

- Maharashtra State Board Class 8 General Science Textbook (Chapter 13: Chemical Change and Chemical Bond).
- Balbharti Standard 8 Science Workbook — Physical vs Chemical Changes.
- Kitchen Chemistry: Experiments for School Students, National Science Centre.